Analysis and Support in Intelligent Cars

Dynamic Modelling for Human-centered Systems

Chapter 15

Summary model-based reasoning

- Domain model → Analysis model:
  - Use meta-statements about domain model
  - Reason from **observations** and **desire** to **assessment**
  - Forward or backward reasoning (or both)
- Domain model → Support model:
  - Extend domain model with support actions
  - Backward reasoning: from **desired state** to **support action**
    - Reasoning about **desires**
  - Forward reasoning: from **assumed action** to **desired effect**
    - Reasoning about (hypothetical) **beliefs**

Overview of today’s lecture

1. **Driving Behaviour - Domain Model**
2. Driving Behaviour - Analysis Model
3. Driving Behaviour - Support Model

Context

- Model-based Intelligent Systems:
  - use sensor information about human functioning
  - reason about the state of the human
  - (re)act by undertaking actions in a knowledgeable manner

Application: Driver Support

Question

- What factors can impair driving behaviour?
- Mention as many factors as possible

Relevant Factors

- Driving behaviour can be impaired by:
  - alcohol intake
  - …
  - …
  - …
  - …
  - …
  - …
  - …
  - …
  - …
  - …
  - …

Relevant Factors

- Driving behaviour can be impaired by:
  - alcohol intake
  - other drug intake (medication, …)
  - fatigue
  - health problems
  - poor eyesight
  - specific emotional states
  - presence of other persons in the car
  - desire for high speed
  - loud music
  - use of mobile phones
  - vehicle factors
  - environment (bad road, no signs, …)

This Lecture:
Driver Support System

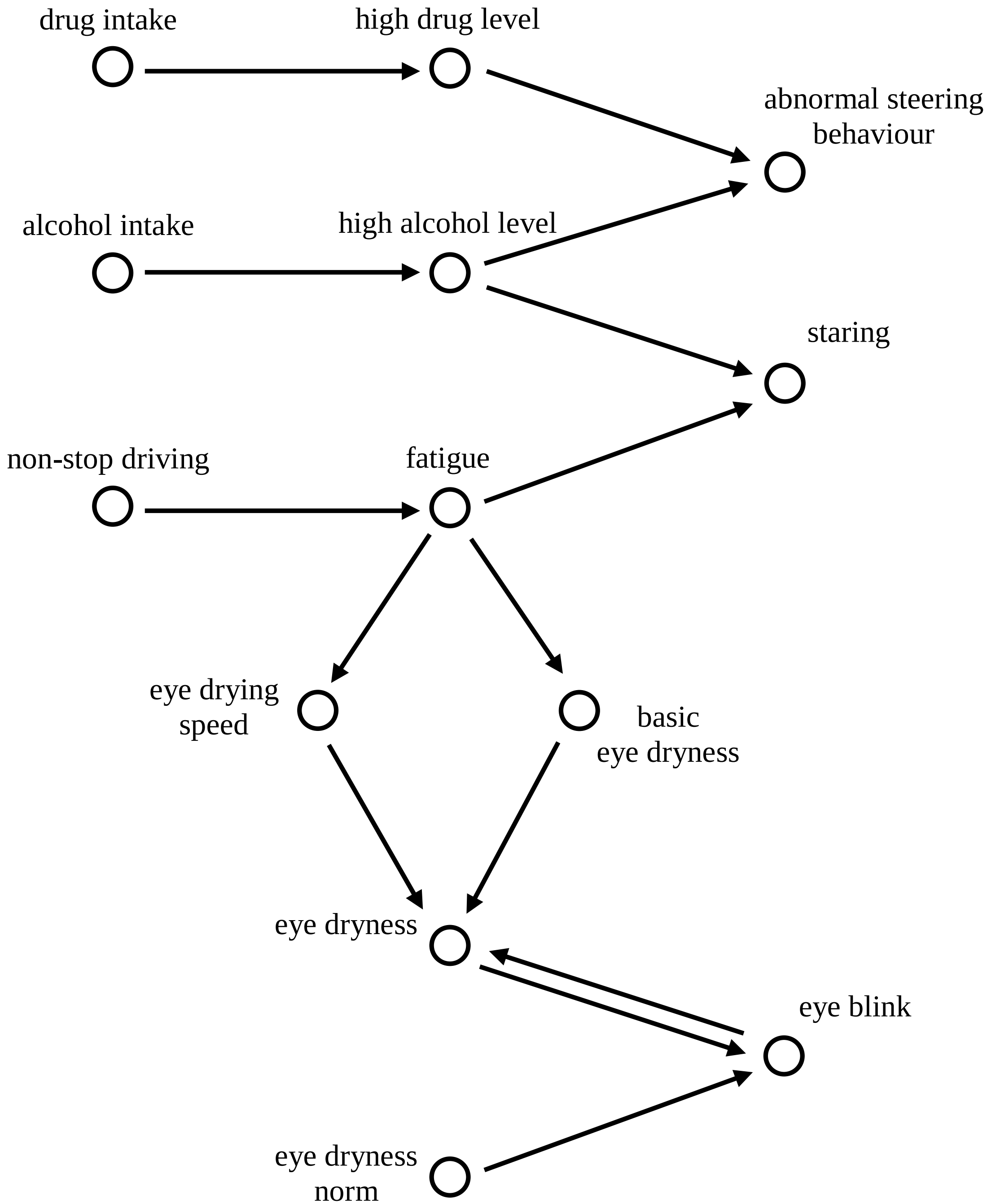
- Improve safety by **monitoring** the driver:
  - steering behaviour
  - sweating (alcohol)
  - gaze and eye movements
- Based on this input, draw conclusions on **driver's state**
- Not capable of driving → provide **support** (e.g. cruise control stops car)

Lexus' Driver Monitoring System

Identification of relevant concepts in domain model

- drug intake
- alcohol intake
- non-stop driving
- drug level (in blood)
- alcohol level (in blood and sweat)
- fatigue
- steering behaviour (more sudden movements)
- staring (when too much alcohol and fatigue)
- eye blinks (more when tired)
- eye dryness
- basic eye dryness (dryer when tired)
- eye drying speed (higher when tired)
- eye dryness norm

Domain Model



Formalisation

concept	*formalisation*
drug intake	consumed(take_drugs)
alcohol intake	consumed(take_alcohol)
non-stop driving	consumed(non_stop_driving)
drug level	has_level(drug_level, L:LEVEL)
alcohol level	has_level(alcohol_level, L:LEVEL)
fatigue	has_level(fatigue, L:LEVEL)
abnormal steering behaviour	performed(abnormal_steering_behaviour)
staring	performed(staring)
eye blinks	performed(eye_blink)
eye dryness	eye_dryness(I:REAL)
basic eye dryness	basic_eye_dryness(I:REAL)
eye drying speed	eye_drying_speed(I:REAL)
eye dryness norm	eye_dryness_norm(I:REAL)
relations between factors and driver states	affects(P:PRODUCT, S:STATE)

Sorts

- Sorts used in predicates:

sort	*description of use*	*elements*
LEVEL	extent to which a certain factor (e.g., drugs, alcohol, or fatigue) is present in a person's body	{low, high}
REAL	the set of real numbers	the set of real numbers
PRODUCT	different products that can be consumed	{drugs, alcohol, non_stop_driving}
STATE	different aspects of the driver's physical state	{drug_level, alcohol_level, fatigue}

- Alternatively, LEVEL could be replaced by REAL

Relations

DDR1 (Consuming a substance)

If a person consumes a certain product (e.g., drugs),
and this product affects a certain aspect of the driver's state (e.g., drug level in his blood),
then this aspect will be high (e.g., (s)he will have a high concentration of drugs in the blood)

consumed(P) & affects(P,S)
$\twoheadrightarrow$ has_level(S, high)

DDR2 (Not consuming a substance)

If a person does not consume a certain product,
and this product affects a certain aspect of the driver's state,
then this aspect will be low

not(consumed(P)) & affects(P,S)
$\twoheadrightarrow$ has_level(S, low)

Relations - 2

DDR3 (From high drug level to abnormal steering)

If a person has a high drug level,

then (s)he will perform abnormal steering behaviour

has_level(drug_level, high)
$\twoheadrightarrow$ performed(abnormal_steering_behaviour)

DDR4 (From high alcohol level to abnormal steering and staring)

If a person has a high alcohol level,

then (s)he will perform abnormal steering behaviour

and (s)he will stare

has_level(alcohol_level, high)
$\twoheadrightarrow$ performed(abnormal_steering_behaviour) $\wedge$ performed(staring)

DDR5 (From fatigue to staring)

If a person has a high state of fatigue,

then (s)he will stare

has_level(fatigue, high)
$\twoheadrightarrow$ performed(staring)

Relations - 3

DDR6 (Effect of fatique on eye dryness)

If a person has a high state of fatigue,

then the person's basic eye dryness will be d1,

and the person's eye drying speed will be s1,

has_level(fatigue, high)

↠ basic_eye_dryness(d1) & eye_drying_speed(s1)

DDR7 (Basic eye dryness)

If a person has a low state of fatigue,

then the person's basic eye dryness will be d2,

and the person's eye drying speed will be s2,

has_level(fatigue, low)

↠ basic_eye_dryness(d2) & eye_drying_speed(s2)

Relations - 4

DDR8 (Eye blinking)

If a person's eye dryness is X,
and the person's eye dryness norm is Y,
and X > Y,
then the person will perform an eye blink

eye_dryness(X) &
eye_dryness_norm(Y) &
X > Y
$\twoheadrightarrow$ performed(eye_blink)

DDR9 (Effect of blinking on dryness)

If a person performs an eye blink,
and the person's basic eye dryness is d,
then the person's eye dryness will be d

performed(eye_blink) &
basic_eye_dryness(D)
$\twoheadrightarrow$ eye_dryness(D)

Relations - 5

DDR10 (Effect of not blinking on dryness)

If a person does not perform an eye blink,

and the person's eye dryness is X,

and the person's eye drying speed is S,

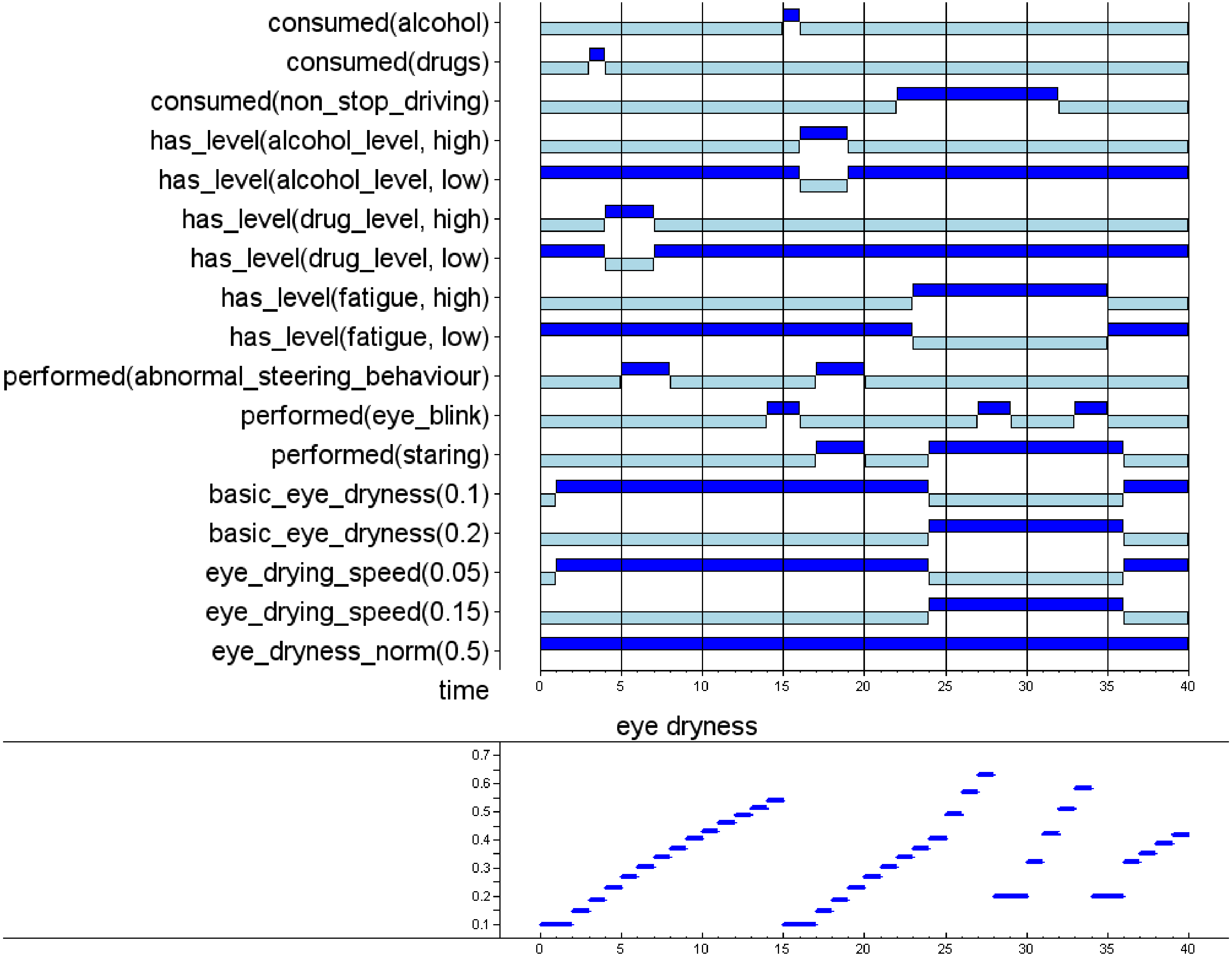
then the person's eye dryness will be X+S*(1-X)

not(performed(eye_blink)) &
eye_dryness(X) &
eye_drying_speed(S)
$\twoheadrightarrow$ eye_dryness(X+S*(1-X))

Simulation

- Settings:
  - in case of low fatigue:
    - basic eye dryness d1=0.1
    - eye drying speed s1=0.05
  - in case of high fatigue:
    - basic eye dryness d2=0.2
    - eye drying speed s2=0.15
  - eye_dryness_norm=0.5

Simulation result



Overview of today's lecture

1. Driving Behaviour - Domain Model
2. **Driving Behaviour - Analysis Model**
3. Driving Behaviour - Support Model

Model for analysis

- Basic idea:
  - intelligent agent should reason about the domain
  - some concepts can be **observed**
    - non-stop_driving
    - eye_blinks
  - some concepts have a **believed** status
    - fatigue
    - eye_dryness, …
  - for some concepts the agent has a specific **desire**
    - fatigue
  - some concepts are **new**
    - assessment on fatigue_discrepancy
      (difference between desired and believed level of fatigue)

Domain Model → Analysis Model

1. Decide which concepts of domain model can be observed
2. Introduce observations for these concepts
3. Decide which concept of domain model is (not) desired
4. Introduce a desire for this concept
5. Introduce beliefs for all concepts
6. Draw arrows from observations via beliefs to the desire (based on forward and/or backward reasoning)
7. Introduce an assessment for the concept from step 3
8. Draw arrows from the desire+belief to the assessment

Domain Model → Analysis Model

1. Decide which concepts of domain model can be observed
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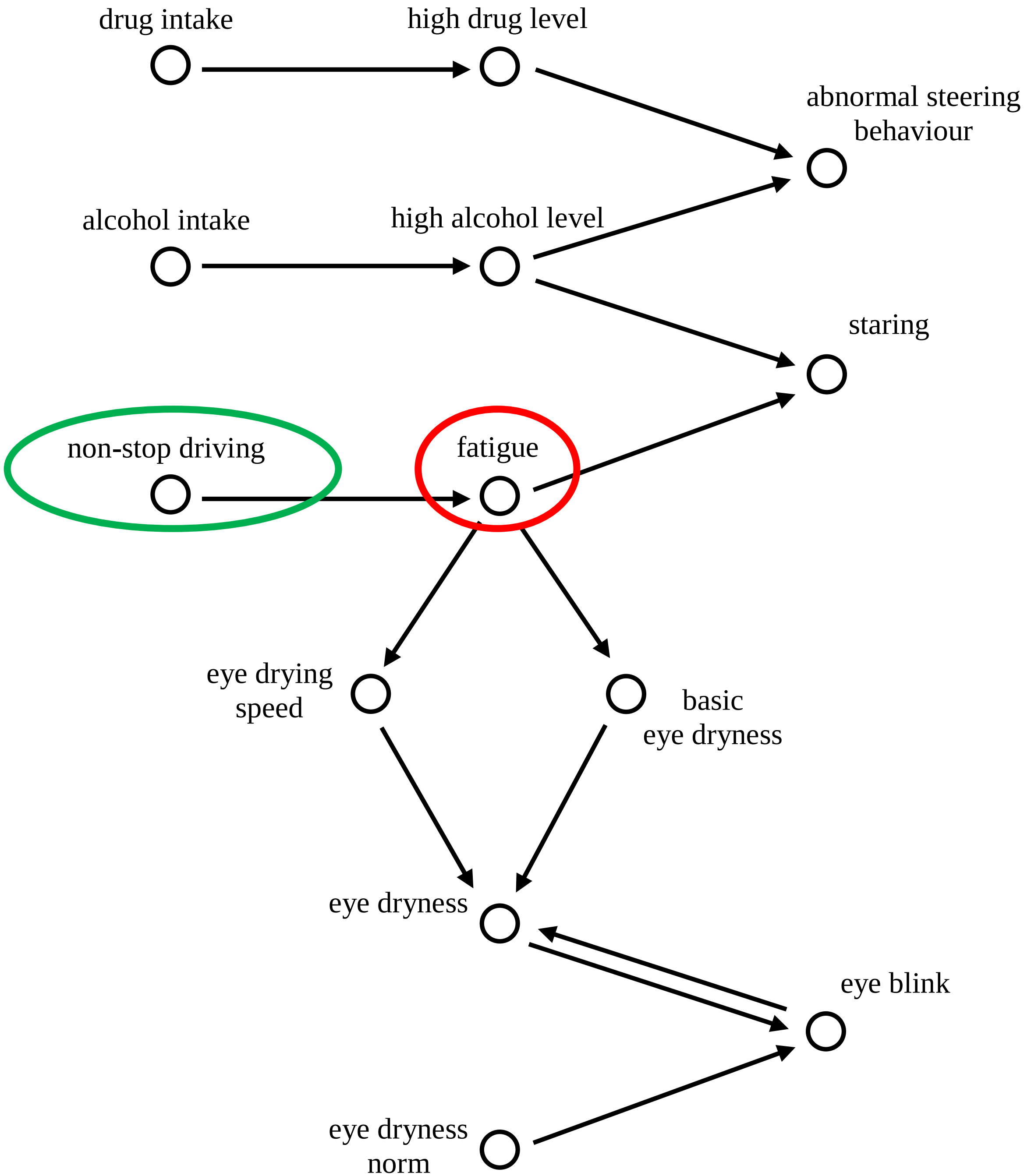
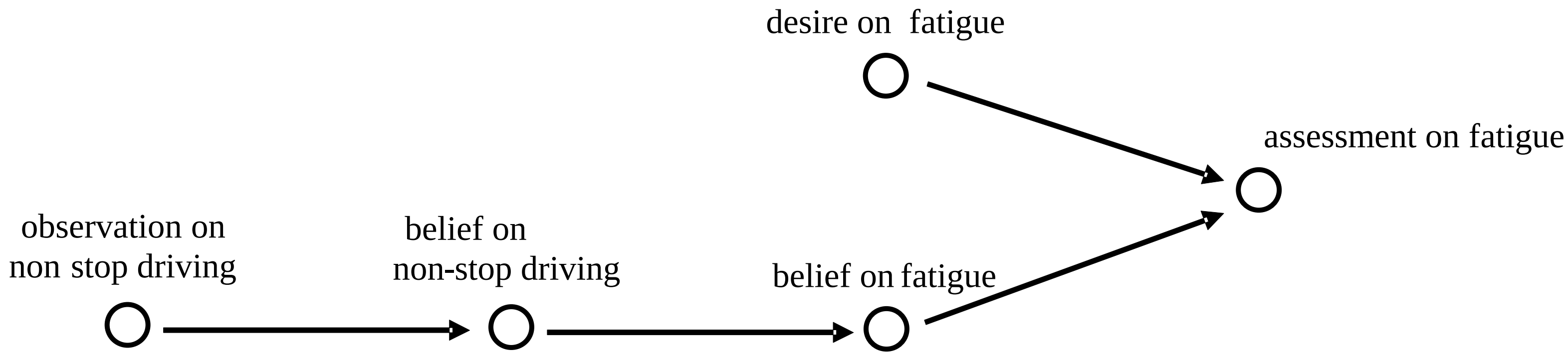
Driver

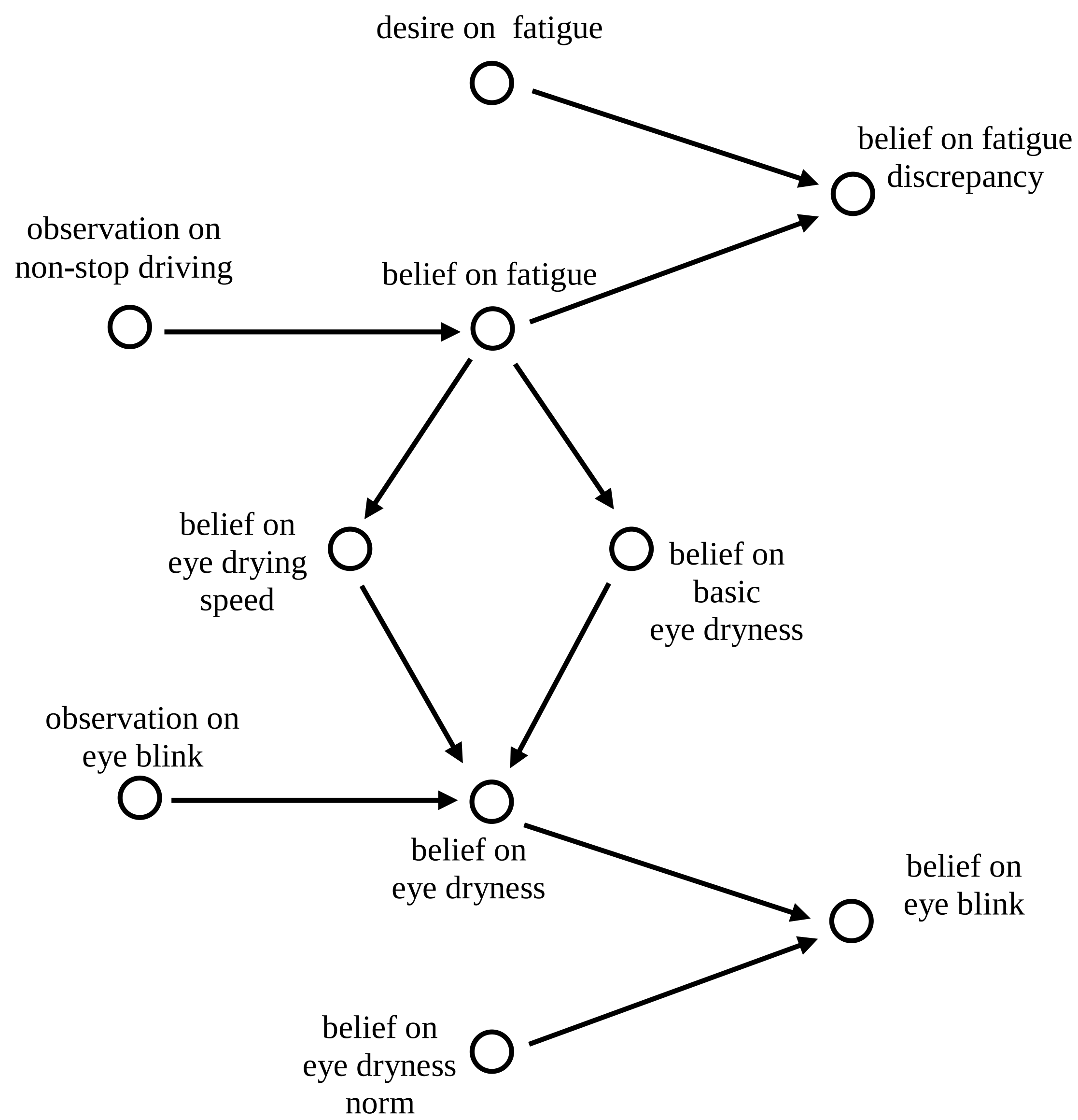


drug intake
high drug level
abnormal steering
behaviour
alcohol intake
high alcohol level
staring
non-stop driving
fatigue
eye drying
speed
basic
eye dryness
eye dryness
eye blink
eye dryness
norm

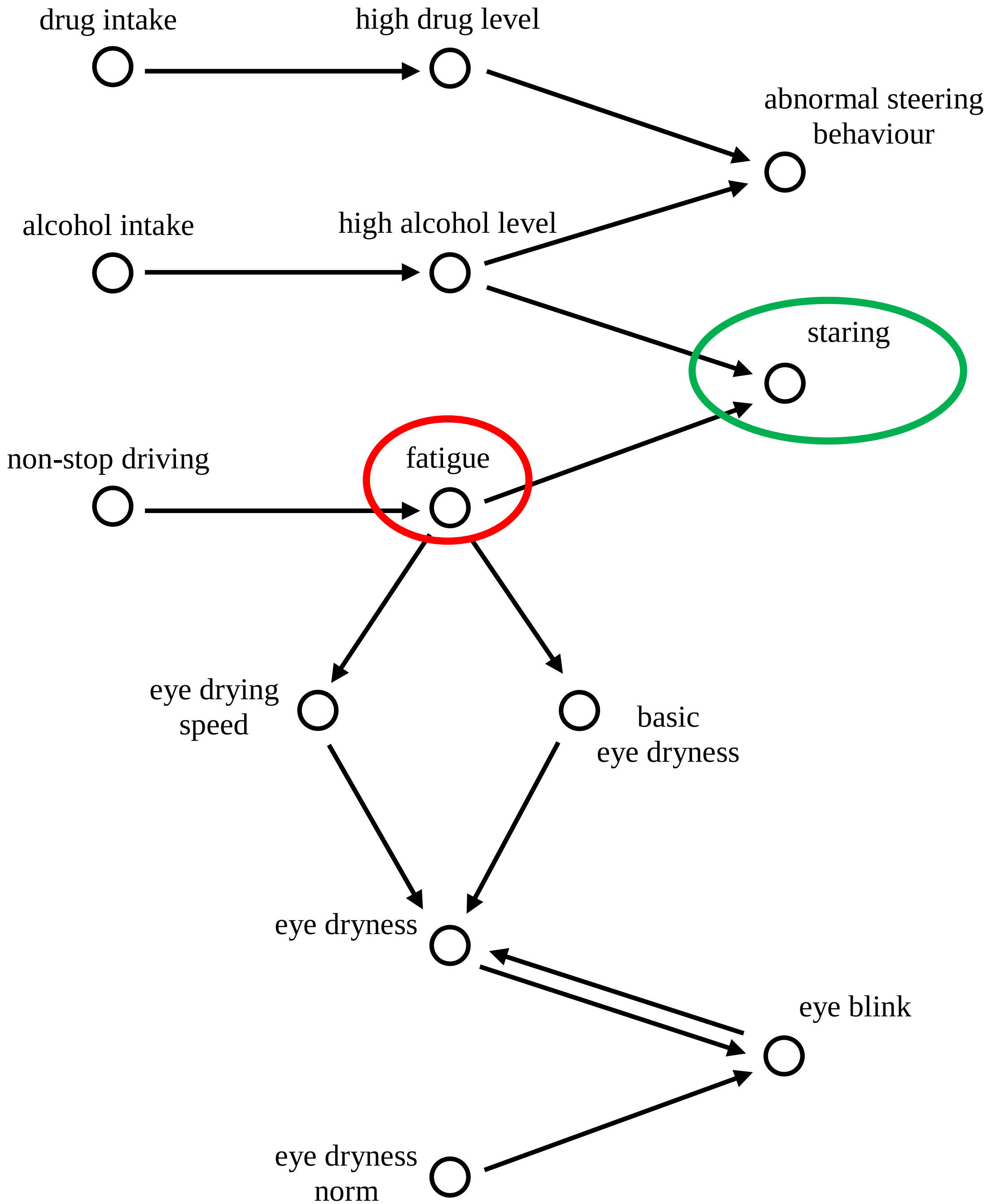
Analysis Model (forward)



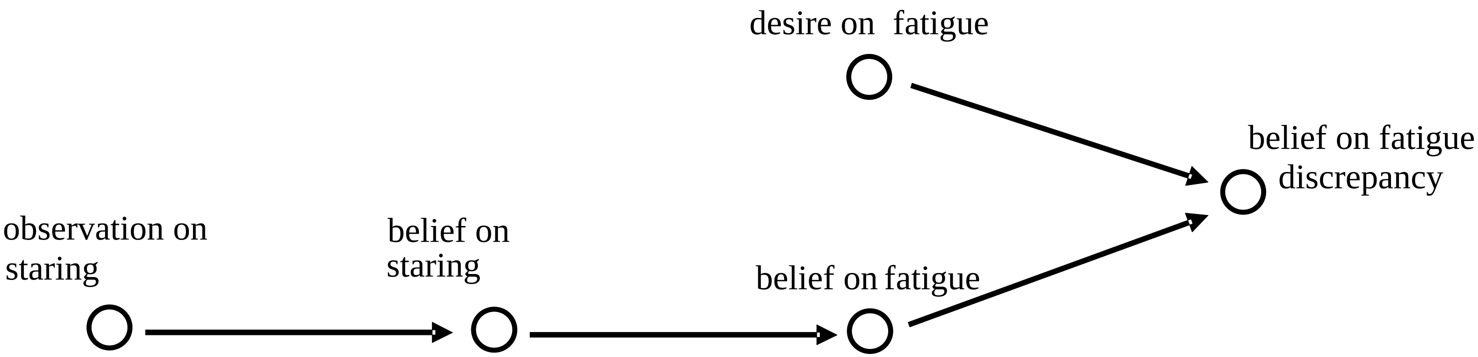
Analysis Model (extended)



Domain


drug intake
high drug level
abnormal steering behaviour
alcohol intake
high alcohol level
staring
non-stop driving
fatigue
eye drying speed
basic eye dryness
eye dryness
eye blink
eye dryness norm

Analysis Model (backward)



Detailed analysis model

concept	*formalisation*
observations on non-stop driving	observed(agent, consumed(non_stop_driving))
observations on eye blinks	observed(agent, performed(eye_blink))
beliefs on fatigue	belief(agent, has_level(fatigue, L:LEVEL))
beliefs on eye blinks	belief(agent, performed(eye_blink))
beliefs on eye dryness	belief(agent, eye_dryness(I:REAL))
beliefs on basic eye dryness	belief(agent, basic_eye_dryness(I:REAL))
beliefs on eye drying speed	belief(agent, eye_drying_speed(I:REAL))
beliefs on eye dryness norm	belief(agent, eye_dryness_norm(I:REAL))
desires on fatigue	desire(agent, has_level(fatigue, L:LEVEL))
beliefs on fatigue discrepancy	belief(agent, has_discrepancy(fatigue, L:LEVEL))
beliefs on relations between factors and driver states	belief(agent, affects(P:PRODUCT, S:STATE))

Relations analysis model - forward

ADR1 (Generating beliefs for consuming a substance)

if the agent observes that a person consumes a certain product (e.g., non-stop driving),
and the agent believes that this product affects a certain aspect of the driver's state (e.g., fatigue),
then the agent will believe that this aspect will be high

observed(agent, consumed(P)) $\wedge$ belief(agent, affects(P,S))
$\twoheadrightarrow$ belief(agent, has_level(S, high))

ADR2 (Generating beliefs for not consuming a substance)

if the agent observes that a person does not consume a certain product,
and the agent believes that this product affects a certain aspect of the driver's state,
then the agent will believe that this aspect will be low

not(observed(agent, consumed(P))) $\wedge$ belief(agent, affects(P,S))
$\twoheadrightarrow$ belief(agent, has_level(S, low))

Relations analysis model - 2

ADR3 (Generating beliefs for effect of high fatique on eye dryness)

if the agent believes that a person has a high state of fatigue,

then the agent believes that the person's basic eye dryness will be d1,

and the agent will believe that the person's eye drying speed will be s1,

belief(agent, has_level(fatigue, high))

$\twoheadrightarrow$ belief(agent, basic_eye_dryness(d1)) $\wedge$ belief(agent, eye_drying_speed(s1))

ADR4 (Generating beliefs for effect of low fatique on eye dryness)

if the agent believes that a person has a low state of fatigue,

then the agent believes that the person's basic eye dryness will be d2,

and the agent will believe that the person's eye drying speed will be s2,

belief(agent, has_level(fatigue, low))

$\twoheadrightarrow$ belief(agent, basic_eye_dryness(d2)) $\wedge$ belief(agent, eye_drying_speed(s2))

Relations analysis model - 3

ADR5 (Generating beliefs for eye blinking)

if the agent believes that a person's eye dryness is X,
and the agent believes that the person's eye dryness norm is Y,
and X > Y,
then the agent will believe (predicts) that the person will perform an eye blink

$$\text{belief(agent, eye\_dryness(X))} \wedge \text{belief(agent, eye\_dryness\_norm(Y))} \wedge x > y$$
$$\twoheadrightarrow \text{belief(agent, performed(eye\_blink))}$$

ADR6 (Generating beliefs for eye dryness)

if the agent observes that a person performs an eye blink,
and the agent believes that the person's basic eye dryness is d,
then the agent will believe that the person's eye dryness will be d

$$\text{observed(agent, performed(eye\_blink))} \wedge \text{belief(agent, basic\_eye\_dryness(D))}$$
$$\twoheadrightarrow \text{belief(agent, eye\_dryness(D))}$$

Relations analysis model - 4

ADR7 (Generating beliefs for effect of not blinking on dryness)

if the agent observes that a person does not perform an eye blink,

and the agent believes that the person's eye dryness is X,

and the agent believes that the person's eye drying speed is S,

then the agent will believe that the person's eye dryness will be X+S*(1-X)

not(observed(agent, performed(eye_blink))) ∧ belief(agent, eye_dryness(X)) ∧ belief(agent, eye_drying_speed(S))

↠ belief(agent, eye_dryness(X+S*(1-X)))

ADR8 (Assessing fatigue)

if the agent desires that the driver has a low state of fatigue,

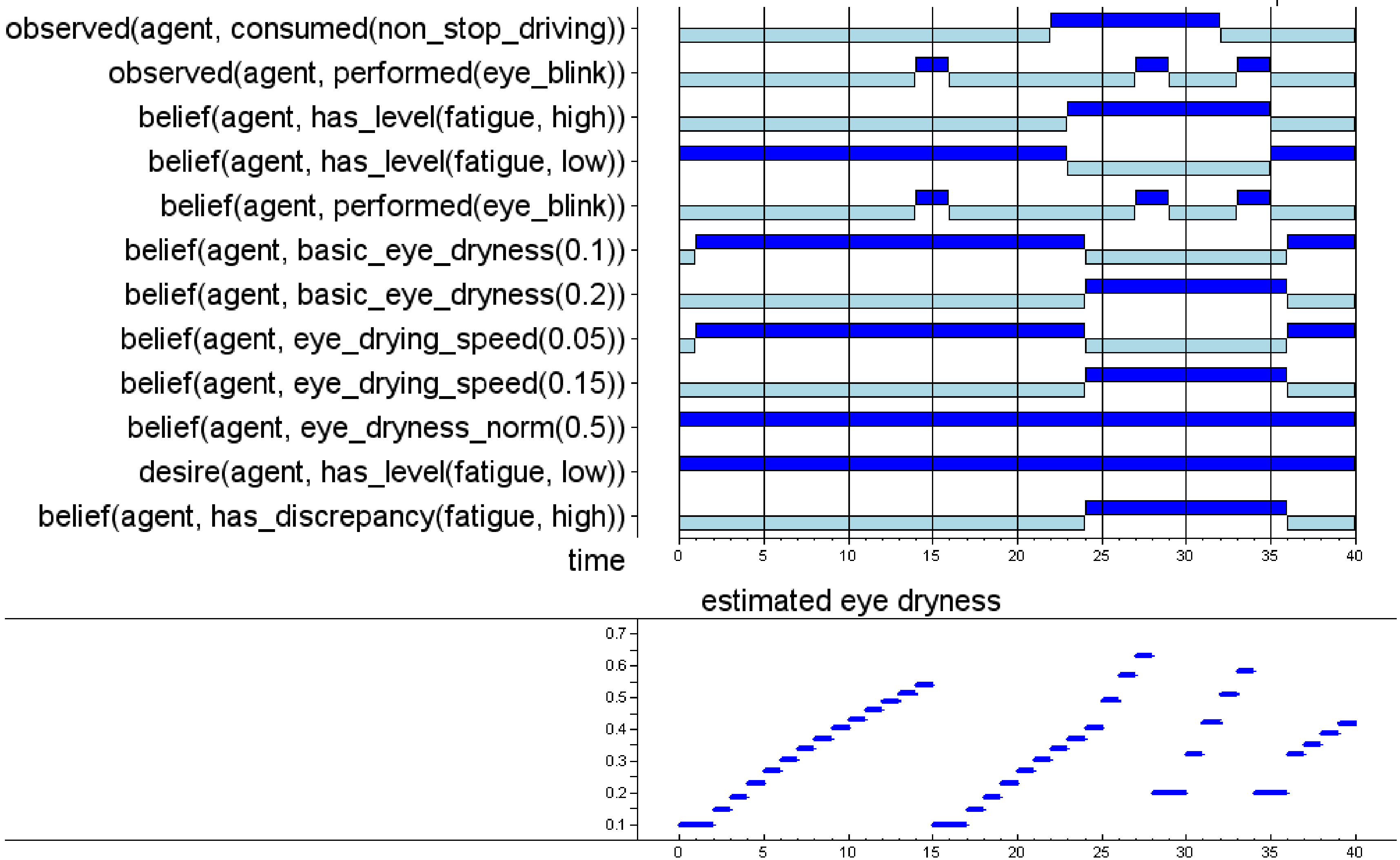
and the agent believes that the driver has a high state of fatigue,

then the agent will believe that the there is a (high) discrepancy between the driver's desired and believed state of fatigue

desire(agent, has_level(fatigue, low)) ∧ belief(agent, has_level(fatigue, high))

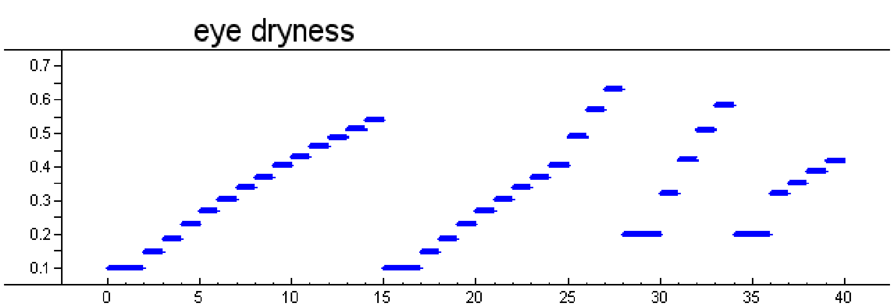
↠ belief(agent, has_discrepancy(fatigue, high))

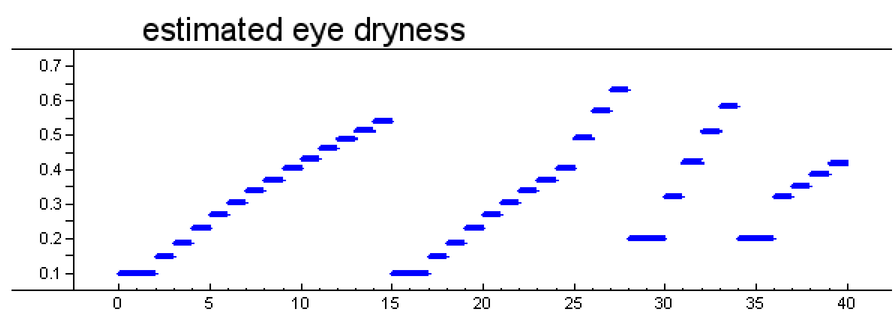
Simulation result



Domain vs. Analysis Model

- Predictions on eye blinks are **100% correct**
  - no adaptation of parameters needed
- In reality, there is usually a **discrepancy** between the model's prediction and reality
  - this is where **parameter adaptation models** come in!





Overview of today's lecture

1. Driving Behaviour - Domain Model
2. Driving Behaviour - Analysis Model
3. **Driving Behaviour - Support Model**

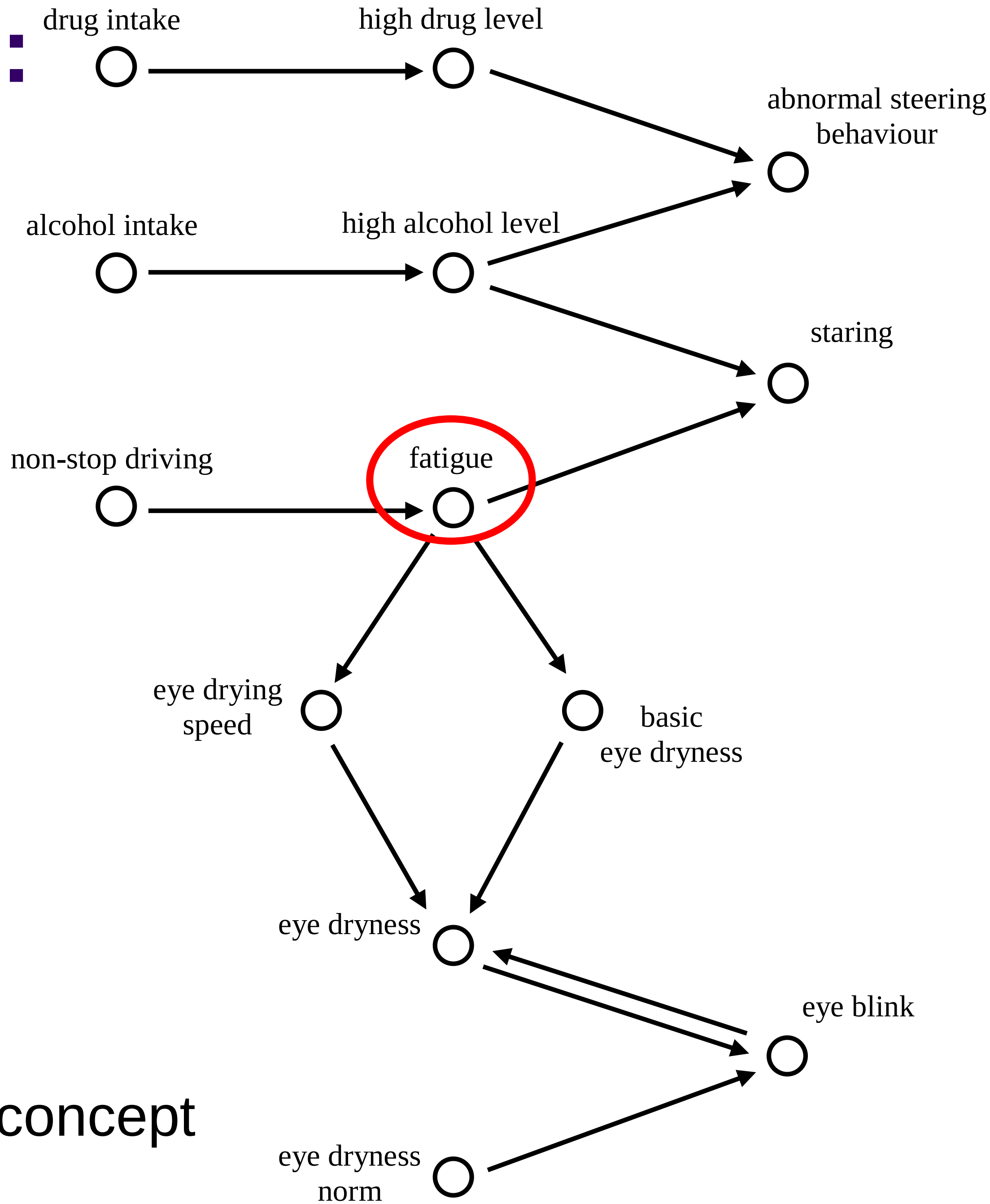
Question

- What kinds of measurements can be taken?
- List 3 measurements
- Groups of 2-3

Model for supporting the driver

- First question: **what kind of measures can be taken?**
  - keep a safe distance with the car in front
  - slow down the car and safely pull it over
  - wake up a sleepy driver
    (e.g., by making a loud sound)
  - give advice to the driver
    (e.g., tell him or her to take a break)
  - …
- Second question:
  when should which measure be taken?

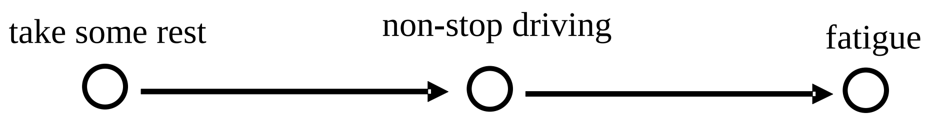
Example: Driver



Decide which concept is not desired

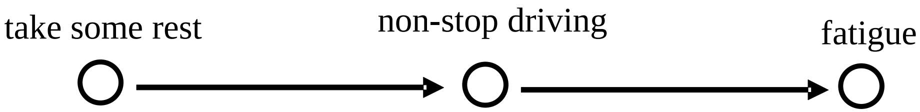
Extended Domain Model

Include support action



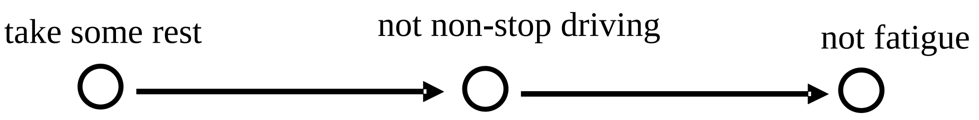
Extended Domain Model

Propagate and remove negative effects



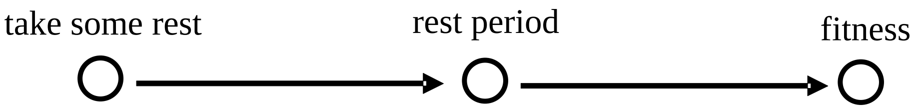
Extended Domain Model

Propagate and remove negative effects



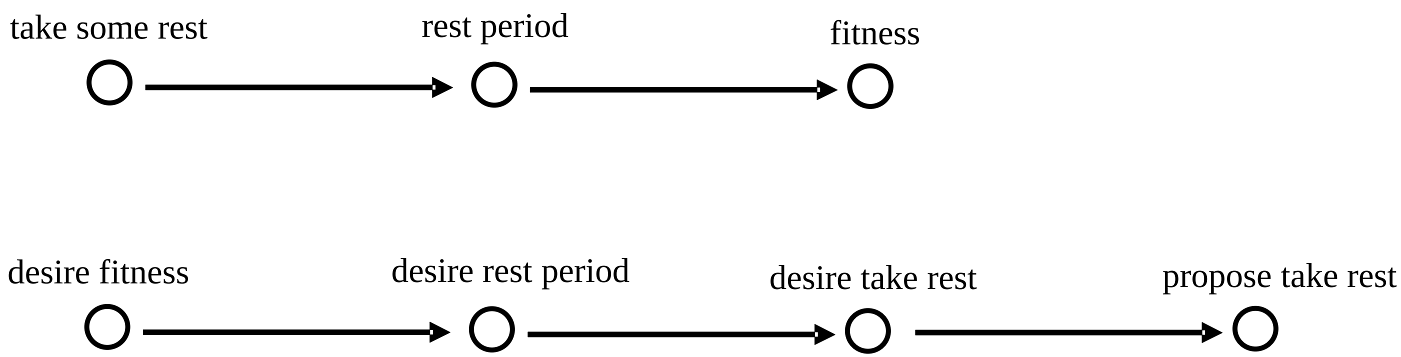
Extended Domain Model

Propagate and remove negative effects



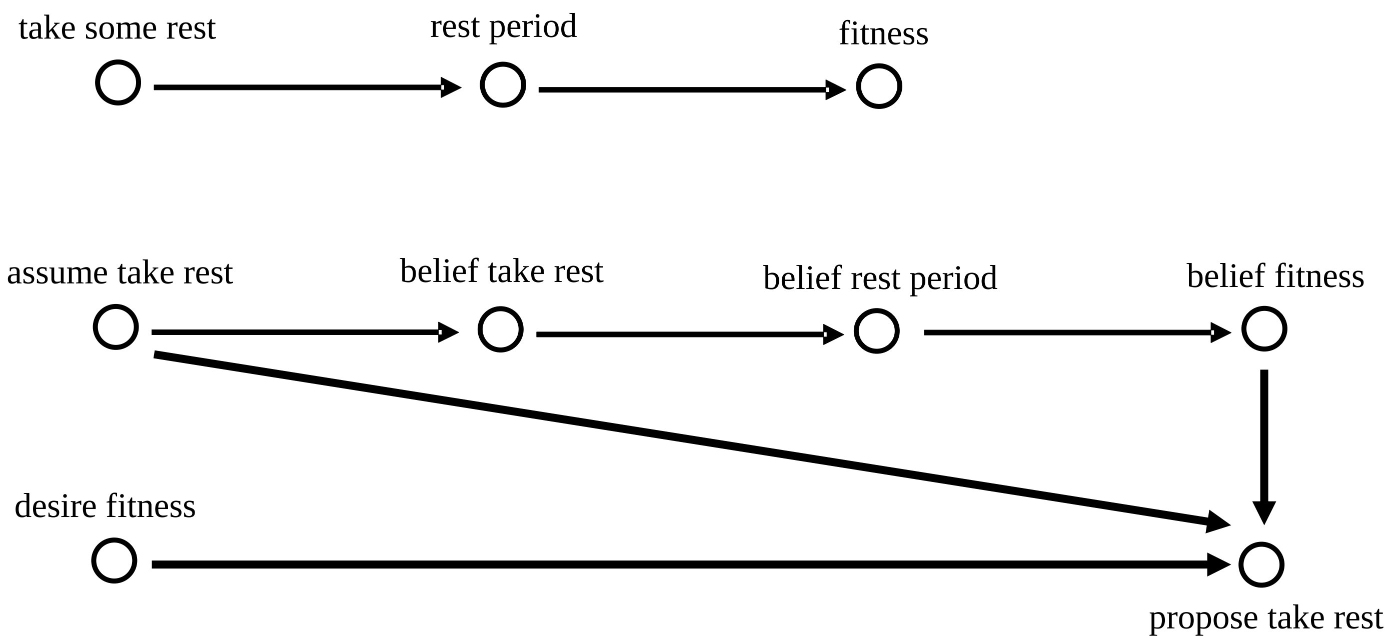
Support Model (backward)

Propose action



Support Model (forward)

support action



Possible extensions

- Taking **more factors** into account than alcohol, drugs, and fatigue
  - emotions, other passengers, music, mobile phones
- Modelling the concentration of alcohol and drugs in more detail, using a continuous, **numerical** approach
- Developing more **complex models for analysis**
  - start from eye-blinking
  - a model to analyse steering patterns by taking speed and direction of steering movements into account
- Considering different and more **complex types of support**
  - a mechanism to compare different options and select the one with the highest feasibility

Summary

- **Intelligent cars** monitor and support the driver
- We first describe the **domain model**
- Then we create an **analysis** model
  - Reasons over domain model using beliefs
- Finally we develop the **support** model
  - Provides interventions when needed